In this paper, we review in typical spaces in the general topology, from the Language of functions.

「Def.

Given a set X and a family of Topological space $\{Y_i\}_{i\in I}$ with functions

$$\{f_i : X \to Y_i\},$$

the topology $\mathcal{U}$ on X is called the initial topology if

$(X, \mathcal{U})$ is the Coarest Topology such that for all $i \in I$, f_i is Continuous

Equivalently,

$$\mathcal{U} = \bigcap \{\mathcal{U}' \mid \text{for all } i\in I,\ f_i : (X, \mathcal{U}') \to Y_i \text{ is continuous}\},$$

and generated by the subbasis $\mathcal{S} := \{f_i^{-1}[U] \mid \text{for some } i \in I,\ U \subseteq Y_i \text{ open}\}$.

Dually, a set X and a family of Topological space $\{Y_i\}_{i\in I}$ with functions

$$\{g_i : Y_i \to X\}$$

the topology $\mathcal{U}$ on X is called the final topology if

$(X, \mathcal{U})$ is the finest Topology such that for all $i \in I$, g_i is Continuous

Equivalently, the $\mathcal{U}$ is final topology if and only if $\mathcal{U}$ satisfies

$U \subseteq X$ is open set if and only if for each $i \in I$, $g_i^{-1}[U] \subseteq Y_i$ is open.

Moreover, given topological space Z and a map $h : Z \to X$, and $w : X \to Z$,

$h : Z \to (X, \mathcal{U}_{init})$ is continuous if and only if $f_i \circ h : Z \to Y_i$ is continuous, for all $i \in I$

$w : (X, \mathcal{U}_{final}) \to Z$ is continuous if and only if $w \circ g_i : Y_i \to Z$ is continuous for all $i \in I$.

Diagramically,
$$\begin{tikzcd} Z \arrow[r, shift left, "h"] & X \arrow[l, shift left, "w"] \arrow[r, shift left, "f_i"] & Y_i \arrow[l, shift left, "g_i"] \end{tikzcd}$$

Product Space

「Sub space

「Metric space

Let X be a set with a Metric $d: X\times X \to \mathbb{R}$.

The Topology $\mathcal{T}_d$ generated by a basis $\{B_r(x) \mid x\in X, r>0\}$ is the coarset topology such that the metric d is continuous.

Prof. Given a basis element $(a,b)\subset\mathbb{R}$, $d^{-1}[(a,b)] = \{(x,y)\in X\times X \mid d(x,y)\in(a,b)\}$.
If $b\le 0$, then the preimage by d is empty, so open. Now, assume that $b>0$.
Let $(x,y)\in d^{-1}[(a,b)]$ be given. Then, $a<d(x,y)<b$.
take $r_x, r_y>0$ such that $r_x+r_y<\min\{d(x,y)-a, b-d(x,y)\}$.
(It is possible, rigorously, take $r_x=r_y=\frac{1}{4}\cdot\min\{d(x,y)-a, b-d(x,y)\}$.).
Then, given $(x',y')\in B_{r_x}(x)\times B_{r_y}(y)$, that is, $d(x,x')<r_x$ and $d(y,y')<r_y$,
By the triangle inequality gives that

$$d(x',y')\le d(x',x)+d(x,y)+d(y,y')<r_x+d(x,y)+r_y<b$$

and

$$d(x,y)\le d(x,x')+d(x',y')+d(y',y)<r_x+d(x',y')+r_y$$

$\Rightarrow a<d(x,y)-(r_x+r_y)<d(x',y')$, so that

$a<d(x',y')<b$, it means $(x',y')\in d^{-1}[(a,b)]$.

(We also used the symmetricity of metric), therefore

$$(x,y)\in B_{r_x}(x)\times B_{r_y}(y)\subseteq d^{-1}[(a,b)].$$

Since $B_{r_x}(x)\times B_{r_y}(y)$ is open set of $X\times X$, so $d^{-1}[(a,b)]$ is open. $\rfloor$ d is continuous.

To show that the second assertion, let $\mathcal{T}'$ be a topology on X such that the Metric $d: X\times X\to\mathbb{R}$ is continuous. Let $x_0\in X$ be fixed.
Since every continuous map is continuous in each variable separately, the map $f_{x_0}: X\to\mathbb{R};\ x\mapsto d(x,x_0)$ is continuous. We want to show that: $\mathcal{T}_d\subseteq\mathcal{T}'$.
Let $B_r(x)\in\mathcal{T}_d$ be an open set of the metric space. Then, for f_x,

$$B_r(x)=f_x^{-1}[(-\infty,r)]$$

is open set of $\mathcal{T}'$, since f_x is continuous on $\mathcal{T}'$ and $(-\infty,r)\subseteq\mathbb{R}$ is open set. $\boxed{/\!/}$

the initial topology with respect to $\{ d_x : X \to \mathbb{R} : y \mapsto d(x, y) \mid x \in X \}$.